

HW 7

1) This ~~is~~ ~~mult. (1, g, g)~~

~~$x_i | y_i = k \sim N(\mu_k, \Sigma)$~~

~~for~~ $y \sim \text{Mult.}(1, (\pi_1, \dots, \pi_K))$

$$\Sigma = \begin{pmatrix} \pi_1(1-\pi_1) & -\pi_1\pi_2 & \dots & -\pi_1\pi_{K-1} \\ \vdots & & & \\ -\pi_1\pi_{K-1} & & & \pi_{K-1}(1-\pi_{K-1}) \end{pmatrix}$$

from KxBook

$$\log L(\beta) = \sum_{i=1}^n \left[\sum_{k=1}^K 1(y_i=k) x_i^T \beta_k - \log \sum_{k=1}^K e^{x_i^T \beta_k} \right]$$

show $\frac{\partial^2 \log L(\beta)}{\partial \beta \partial \beta^T}$ is ~~negative~~ ~~semidef.~~

negative semi def $\rightarrow$ symm. matrix w/ all eigenvalues non positive

$$\frac{\partial^2 \log L(\beta)}{\partial \beta \partial \beta^T} = \begin{bmatrix} \end{bmatrix} \in \mathbb{R}^{p(K-1) \times p(K-1)}$$

Goal:

quadratic form

$x^T A x \leq 0$ for all vectors of matrix

$\nwarrow$ no less than

for the $n \times n$ Hessian matrix H

for the Hessian matrix H ;

$$v^T H v = \sum_{k=1}^{K-1} \sum_{l=1}^{K-1} v_k^T H v_l$$

$$= - \sum_{k,l} \Sigma_{kl} \underbrace{(v_k^T x)}_{\text{some scalar}} \underbrace{(x^T v_l)}_{\text{some scalar}}$$

since Σ_{kl} is positive semi-definite

(from proposition 21.1)

then the Hessian H is negative semi-definite

2) assume $y_i^* | x_i \sim \text{pois}(e^{x_i^T \beta})$

$$y_i = 1(y_i^* > 0)$$

show $y_i | x_i$ follows cloglog st

$$P(y_i = 1 | x_i) = g(x_i^T \beta)$$

where $g(z) = 1 - \exp(-e^z)$

$$P(y_i = 1 | x_i) = P(y_i^* > 0 | x_i)$$



$$P(y_i^* > 0 | x_i) = 1 - P(y_i = 0 | x_i)$$

Given Poiss ; $P(y_i = 0) = \frac{e^{-\lambda_i} \lambda_i^0}{0!} = e^{-\lambda_i}$

$$P(y_i^* > 0 | x_i) = P(y_i = 1 | x_i) = 1 - e^{-\lambda_i}$$

//
 $g(z) = 1 - \exp(-e^z)$

3) derive cluster cov. matrix for Poisson

in Poisson regression;
log likelihood is

$$li(\beta) = \sum \left[y x^T \beta - e^{x^T \beta} - \log(y!) \right]$$

~~note~~ solve for

using
matrix
calculus
remember
matrix?

$$\frac{-2}{\partial \beta^T} = \sum e^{x_i^T \beta} x_i + x_i^T = x^T \text{diag}\{e^{x_i^T \beta}\} x$$

$$\Sigma \text{Var}(si(\beta_0)) = \sum x_i^T E[\varepsilon_i \varepsilon_i^T] x_i$$

$$\widehat{\text{cov}}(\hat{\beta}) = \left(\frac{-2}{\partial \beta^T} \right)^{-1} \left(\Sigma \text{Var}(si(\beta_0)) \right) \left(\frac{-2}{\partial \beta^T} \right)^{-1}$$

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4. a)

```
treaty <- read.csv("treaty.csv")

covs <- c("shift_left", "flexible", "regnorm", "gdpgrow",
          "resgdp", "bopgdp", "useimfcr", "surveil",
          "univers", "resvol", "totvol", "tradedep",
          "military", "termlim", "parli", "lastrest",
          "lastrest2", "lastrest3")

form <- as.formula(
  paste("restrict ~ art8 +", paste(covs, collapse = " + "))
)

model <- glm(form, data = treaty, family = binomial)

summary(model)
```

Call:

```
glm(formula = form, family = binomial, data = treaty)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.2730042	0.6012649	0.454	0.649793
art8	-1.6955541	0.1621335	-10.458	< 2e-16 ***
shift_left	0.6228087	0.2470776	2.521	0.011712 *
flexible	-0.5891436	0.1465208	-4.021	5.80e-05 ***
regnorm	0.0045765	0.0023533	1.945	0.051803 .


```

gdpgrow      -0.0101136  0.0054561  -1.854  0.063792  .
resgdp       -0.4410251  0.3948874  -1.117  0.264064
bopgdp       -0.0031516  0.0053319  -0.591  0.554466
useimfcr     0.9478057  0.1282076   7.393  1.44e-13 ***
surveil      0.4333229  0.1584113   2.735  0.006230 **
univers      0.0362446  0.0063689   5.691  1.26e-08 ***
resvol       0.1398819  0.1103494   1.268  0.204931
totvol       0.3661580  0.0966663   3.788  0.000152 ***
tradedep     -0.0063087  0.0013369  -4.719  2.37e-06 ***
military     -0.3203585  0.1400044  -2.288  0.022126 *
termlim      0.0483649  0.1487620   0.325  0.745093
parli        -0.0764728  0.1527916  -0.501  0.616720
lastrest     -1.6722703  0.0746347 -22.406  < 2e-16 ***
lastrest2    0.1344814  0.0089264  15.066  < 2e-16 ***
lastrest3    -0.0030465  0.0002726 -11.174  < 2e-16 ***

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

```

Null deviance: 6013.5  on 4361  degrees of freedom
Residual deviance: 2261.3  on 4342  degrees of freedom
AIC: 2301.3

```

Number of Fisher Scoring iterations: 7

This is an observational regression, not a truly randomized experiment. Even with controlling for covariates, there are omitted variable bias. This result reflects association not causal effect.

b)

```

library(sandwich)
library(lmtest)

```

Loading required package: zoo

Warning: package 'zoo' was built under R version 4.5.2

Attaching package: 'zoo'

The following objects are masked from 'package:base':

as.Date, as.Date.numeric

```
cluster_se <- vcovCL(model, cluster = ~ imf_ccode)
coeftest(model, vcov = cluster_se)
```

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.27300418	0.83175212	0.3282	0.7427394	
art8	-1.69555413	0.24731223	-6.8559	7.085e-12	***
shift_left	0.62280870	0.21728378	2.8663	0.0041525	**
flexible	-0.58914357	0.18433963	-3.1960	0.0013936	**
regnorm	0.00457654	0.00365558	1.2519	0.2105940	
gdpgrow	-0.01011359	0.00598992	-1.6884	0.0913278	.
resgdp	-0.44102511	0.73503204	-0.6000	0.5485009	
bopgdp	-0.00315159	0.00506683	-0.6220	0.5339392	
useimfcr	0.94780573	0.16801678	5.6411	1.689e-08	***
surveil	0.43332288	0.21325312	2.0320	0.0421572	*
univers	0.03624463	0.00953594	3.8008	0.0001442	***
resvol	0.13988193	0.16009121	0.8738	0.3822468	
totvol	0.36615803	0.11892074	3.0790	0.0020769	**
tradedep	-0.00630870	0.00175369	-3.5974	0.0003214	***
military	-0.32035852	0.15113387	-2.1197	0.0340313	*
termlim	0.04836490	0.18107791	0.2671	0.7893965	
parli	-0.07647276	0.16518569	-0.4630	0.6434000	
lastrest	-1.67227028	0.10391680	-16.0924	< 2.2e-16	***
lastrest2	0.13448139	0.01613614	8.3342	< 2.2e-16	***
lastrest3	-0.00304646	0.00056163	-5.4243	5.818e-08	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Clustering should make the evaluation more robust, but no significant change is observed.

c)

Ridge regression typically shrinks the coefficients towards 0 (with the goal of increasing prediction stability). The goal with this eval is to conduct inference on the art8 coefficient, so utilizing ridge regression is not ideal.